8 hours on examples (boy it took *long*)

Problem 1 (Z3EOED6A)

We claim that, if $a \leq b \leq c$,

$$(a+c) \cdot b \geq a \cdot c + b^2$$

We may let $b = a + x; c = b + y$ for non-negative x, y.

$$\Leftrightarrow (2a + x + y) \cdot (a + x) \geq a \cdot (a + x + y) + (a + x)^2$$

$$\Leftrightarrow 2a^2 + 3ax + ay + x^2 + xy \geq a^2 + ax + ay + a^2 + 2ax + x^2$$

$$\Leftrightarrow xy \geq 0$$

Using the claim,

$$x_1 x_{100} + x_2 x_{99} + \cdots + x_{50} x_{51} \leq -50 \cdot x_{50}^2$$

$$\Leftrightarrow \left(x_1 x_{100} + x_{50}^2\right) + \left(x_2 x_{99} + x_{50}^2\right) + \cdots + \left(x_{50} x_{51} + x_{50}^2\right) \leq 0$$

$$\Leftarrow x_{50}(x_1 + x_{100} + x_2 + x_{99} + \cdots) \leq 0$$

$$\Leftrightarrow 0 \leq 0$$

Problem 3 (16EGMO1)

For the sake of contradiction, let this not be the case.

This would mean $\min\left(x_i^2 + x_{i+1}^2\right) > 2x_j x_{j+1}$ for any index j. By the extreme value theorem, there is an i when $x_i^2 + x_{i+1}^2$ is minimized. WLOG, let $i = 0$ and $x_1 \geq x_0$.

We take indices modulo the length, so x_{-1} is the last term of the array.

Claim: For any integer k, we have $x_{2k}, x_{-2k} \leq x_0 \leq x_1 \leq x_{2k+1}, x_{-2k-1}$. In other words, all the even-indexed terms are less than x_0 while all the odd-indexed terms are greater than x_0.

We prove this by mathematical induction.

Base Case: $n = 0$, $x_0 \leq x_0$ we are done.

Inductive step:

From $2k$ to $2k+1$. We have $x_{2k}, x_{-2k} \leq x_0$. However, we know that $x_0^2 + x_1^2 \leq x_i^2 + x_{i+1}^2$ for any integer i. Therefore,

$$x_0^2 + x_1^2 \leq x_{2k}^2 + x_{2k+1}^2 \Leftrightarrow x_0^2 - x_{2k}^2 \leq x_{2k+1}^2 - x_1^2$$

Since everything is non-negative, $x_0 \geq x_{2k} \Longrightarrow x_{2k+1} \geq x_1$. The case for x_{-2k} is analogous.

The case for $2k+1$ to $2k+2$ is analogous. $x_0^2 + x_1^2 \leq x_{2k+2}^2 + x_{2k+1}^2 \Longrightarrow x_{2k+2} \geq x_0$.

Whenever the external terms x_{-2k}, x_{2k} are both greater than x_1, then note that,

$$2x_{-2k}x_{2k} \geq 2x_1^2 \geq x_0^2 + x_1^2$$

Whenever the external terms x_{-2k-1}, x_{2k+1} are both smaller than x_0, note that,

$$x_{-2k-1}^2 + x_{2k+1}^2 \leq 2x_0^2 \leq x_0^2 + x_1^2$$

Both contradict previous assumptions, so we are done.

6 clubs; 10 hours

Problem 4 (06AMO2)

We claim the answer is $N = k(k+1)(2k+1)$.

This is attained for the sequence,

$$k^2, k^2 + 1, \dots, k^2 + 2k$$

The sum of the terms is $\frac{(2k+1)(k^2+k^2+2k)}{2} = N$. The subset of size k with maximal sum is obviously,

$$\{k^2 + k + 1, \dots, k^2 + 2k\}$$

This subset has sum $\frac{k(2k^2+3k+1)}{2} = \frac{k(2k+1)(k+1)}{2} = \frac{N}{2}$.

To prove this is minimum, we first prove that when N is minimized, the sequence is consecutive.

Let a set $A = \{a_1, a_2, \dots, a_{2k+1}\}$ for $a_1 < a_2 < \cdots < a_{2k+1}$ attain the minimum N. We may have the difference sequence $d_i = a_{i+1} - a_i$ for $1 \leq i \leq 2k$.

FTSOC, let there exist some $d_i > 1$. We have,

$$a_2 = a_1 + d_1;$$

$$a_3 = a_1 + d_1 + d_2;$$

$$a_4 = a_1 + d_1 + d_2 + d_3;$$

$$\dots$$

$$a_{2k+1} = a_1 + d_1 + \cdots + d_{2k}$$

We have,

$$\frac{N}{2} = \frac{1}{2}\sum_{i=1}^{2k+1} a_i \geq \sum_{i=k+2}^{2k+1} a_i$$

With the right-hand side being the sum of the maximum subset of size k, or the sum of the k biggest numbers.

$$\Leftrightarrow \sum_{i=1}^{2k+1} a_i \geq \sum_{i=k+2}^{2k+1} a_i + \sum_{i=k+2}^{2k+1} a_i$$

$$\Leftrightarrow \sum_{i=1}^{k+1} a_1 + d_1 + \cdots + d_{i-1} \geq \sum_{i=k+2}^{2k+1} a_1 + d_1 + \cdots + d_{i-1}$$

Since all d are greater than 1,

$$\Rightarrow \sum_{i=1}^{k+1} a_1 + i - 1 \geq \sum_{i=k+2}^{2k+1} a_1 + i - 1$$

$$\Leftrightarrow \frac{\sum_{i=1}^{2k+1} a_1 + i - 1}{2} \geq \sum_{i=k+2}^{2k+1} a_1 + i - 1$$

Therefore, the set $\{a_1, a_1 + 1, \dots, a_1 + 2k\}$ also has every subset of size k sum to less than half of the total sum. The latter set also has a smaller N, so we have a contradiction.

We now simply find the smallest set $\{x, x + 1 , \dots, x + 2k\}$ following the conditions.

$$\sum_{i=x+k+1}^{x+2k} i \leq \frac{N}{2}$$

$$\Leftrightarrow k(2x + 3k + 1) \leq N = (x + k)(2k + 1)$$

$$\Leftrightarrow 2kx + 3k^2 + k \leq 2xk + 2k^2 + x + k$$

$$\Leftrightarrow x \geq k^2$$

Therefore, the smallest sum is attained when the set is $\{k^2, k^2 + 1, \dots, k^2 + 2k\}$. As seen earlier, this sum is $k(k + 1)(2k + 1)$.

Problem 5 (KAZ)

We claim the answer is 150, attained with 50 groups of $(5, 1, 1, 1, 1)$ and 100 groups of $(5, 2, 2)$.

Let the unit of value be in dollars for convenience.

Note that, at most, we may only use 150 of the coins of value \$5. This is because every time we use a coin of value \$5, we must also use \$4 of \$1 or \$2 coins to get the group of \$9. We only have \$600 of coins of value \$1 and \$2 so we may only make 150 such sets.

Therefore, we cannot use 50 of the coins of \$5 meaning it'd be the same if they never existed. In the problem where we have two-hundred \$1, \$2 coins and one-hundred-fifty \$5 coins, we can only make 150 groups of \$9 with \$1350.

11 clubs; 11.5 hours

Problem 7 (14TWNQ3J5)

We start with proving the lower-bound.

$$\frac{k_1+\cdots+k_n}{n}=\frac{\left(\frac{x_n}{x_1}+\frac{x_2}{x_1}\right)+\left(\frac{x_1}{x_2}+\frac{x_3}{x_2}\right)+\cdots+\left(\frac{x_{n-1}}{x_n}+\frac{x_1}{x_n}\right)}{n}$$

By AM-GM, this quantity is clearly greater than or equal to 2 with equality case when $x_0=\cdots=x_n$.

We claim the maximum $k_1+\cdots+k_n$ for such a sequence is $3n-1$. This is accomplished for example with the sequence, $1,2,3,\dots,n$. Namely, $k_1=n+2; k_2=\cdots=k_{n-1}=2; k_n=1$.

We prove this by mathematical induction.

Base case: $n=1$. Let the only term be x, $k_1=\frac{x+x}{x}=2=3n-1$.

Inductive step: $n=k$ using $n=k-1$

We let $x_1,x_2,\dots,x_n$ be the sequence that maximizes $\sum k_i$. We let x_i be one of the maximum terms in the sequence.

Case 1: $x_i=x_{i-1}$ or $x_i=x_{i+1}$

If $x_i=x_{i-1}$, we see that $x_i \mid x_{i+1}$. However, $0<x_{i+1}\le x_i$ so $x_{i+1}=x_i$. We may recursively do this to find that $x_1=x_2=\cdots=x_n$. Therefore, $k_1=k_2=\cdots=k_n=2$. Since $\sum k_i=2n\le 3n-1$ for positive integers n, we are done. The case for $x_i=x_{i+1}$ is analogous.

Case 2: $x_{i-1}\ne x_i\ne x_{i+1}$

Since $x_i \mid x_{i-1}+x_{i+1}$ but $x_{i-1},x_{i+1}<x_i$, we must have $x_i=x_{i-1}+x_{i+1}$. We let,

$$a=x_{i-1}; a+b=x_i; b=x_{i+1}$$

We now remove x_i from the sequence. Note that $k_1,\dots,k_{i-2}$ stay the same, k_{i-1} decreases by one as its right neighbor is a less than what it was. $k_i=1$ disappears. k_{i+1} decreases by one as its left neighbor is now b less than what it was. $k_{i+2},\dots,k_n$ stay the same. Finally, $k_{i+2},\dots,k_n$ shift positions left by one index.

In the current case, by the induction hypothesis the new $\sum k_i\le 3(n-1)-1$. Since $\sum k_i$ had dropped by 3, we are done.

Conclusion: By induction we're done.

With the claim proven, we have,

$$\frac{k_1+k_2+\cdots+k_n}{n}\le 3-\frac{1}{n}<3$$

Problem 11 (09USATST7)

We claim the only solution is $x=2; y=4; z=6$ which works if you plug in the terms.

We let $X \coloneqq x-2; Y \coloneqq y-4; Z \coloneqq z-6$.

We prove that $X = Y = Z = 0$ is the only solution.

$$x^3 = 3x - 12y + 50$$

$$\Leftrightarrow X^3 + 6X^2 + 12X + 8 = 3X + 6 - 12Y - 48 + 50$$

$$\Leftrightarrow Y = \frac{-X^3 - 6X^2 - 9X}{12}$$

Subsequently,

$$y^3 = 12y + 3z - 2$$

$$\Leftrightarrow Y^3 + 12Y^2 + 48Y + 64 = 12Y + 48 + 3Z + 18 - 2$$

$$\Leftrightarrow Z = \frac{Y^3 + 12Y^2 + 36Y}{3}$$

Finally,

$$z^3 = 27z + 27x$$

$$\Leftrightarrow Z^3 + 18Z^2 + 108Z + 216 = 27Z + 162 + 27X + 54$$

$$\Leftrightarrow X = \frac{Z^3 + 18Z^2 + 81Z}{27}$$

If $X > 0$, then $Y < 0$ so $Z < 0$. This contradicts with the last equation that suggests $X < 0$.

If $X < 0$, then $Y > 0$ so $Z > 0$. This contradicts with the last equation that suggests $X > 0$.

Therefore, $X = 0 \Longrightarrow Y = 0 \Longrightarrow Z = 0$ and we are done.

Problem 13 (16HRVTST22)

WLOG, let the two black squares be the top-left and bottom-right.

We claim the minimum is $2n - 4$.

We color all the squares in the top row and right-most column except for the corners for a total of $(n-2) + (n-2)$ squares colored. We can then flip to top row, then all the columns except for the right-most column to be done.

Note that all moves are reversible, so we do the problem in reverse. We may instead start from a blank state and try to flip colors to create a board where the top-left and bottom-right squares are black.

Note that the order we flip rows or columns do not matter. Also, if we flip any row or column twice, it'd be the same as not having flipped it at all. Therefore, we can let $r_1, r_2, \dots, r_n \in \{0,1\}$ be whether each row is flipped and $c_1, \dots, c_m \in \{0, 1\}$ be whether each column is flipped.

Finally, note that if we swap r_i and r_j, for example, it'd be the same as swapping rows i, j in our resulting graph. Therefore, since we can swap rows and columns, WLOG, we may also say we simply flip the colors first a rows and the first b columns. Note that, even though this swapping means the top-left and bottom-right square may move around, they may never move to the same row or column.

Note that the number of black squares in this case is $(n-a)b + (n-b)a$. We want to minimize the number of black squares. FTSOC, let there be a valid arrangement with less than $2n-2$ squares.

WLOG, let $a \leq b$. By the rearrangement inequality,

$$(n-a)b + (n-b)a \geq (n-a)a + (n-b)b$$

Note that if $1 \leq a \leq n-1$ and $1 \leq b \leq n-1$, then $(n-a)a \geq (n-1)$ and $(n-b)b \geq (n-1)$. This means it's impossible to get less than $2n-2$ here.

If $a = 0$,

$$(n-a)b + (n-b)a = bn < 2n-2 \Longrightarrow b \leq 1$$

However, if we only flip one row, we cannot have the required two black squares in distinct rows and columns so we cannot make the corners black.

The cases where $a = n, b = 0, b = n$ are analogous.

24 clubs; 14 hours

<u>Problem 15 (11RUS93)</u>

We generalize the problem to any n-gon for $n \geq 3$.

We claim the answer is $2n-6$.

Let the vertices be named $1, \dots, n$. We start by drawing the $n-3$ edges,

$$2 \to 4; 3 \to 5; 4 \to 6; \dots; n-2 \to n$$

Each edge after the first only intersects the one before it. Next, we draw the $n-3$ edges,

$$1 \to 3; 1 \to 4; 1 \to 5; \dots; 1 \to n-1$$

The edge $1 \to i$ only intersects one previous edge, namely, $i-1 \to i+1$.

We prove the upper-bound is $2n-6$ using strong induction.

Base case: $n = 3$, no diagonal may be drawn so 0. We are done.

Inductive step: $n = k$ using $n = k-1, \dots, 1$.

Let the last diagonal drawn cut the polygon of size k into two polygons of size $i+1, k-i+1$. We have at most $2(i+1) - 6 + 2(k-i+1) - 6 = 2k-8$ diagonals strictly inside these respective polygons. There can be at most one edge between the polygon since the we just drawn the last edge.

Problem 17 (21SLC2)

We call the colors $c_1, c_2, \dots, c_n$.

We claim the answer is $\frac{(n-1)(n-2)}{2}$.

We claim m works if and only if it is of the form $m = a(n+1) + b(n)$ for some $a, b \in \mathbb{Z}$. If the claim is the numbers are the only ones that are impossible, (I'm wondering if I can just leave this list below without further explanation)

$$n+2, n+3, \dots, 2n-1,$$

$$2n+3, 2n+4, \dots, 3n-1,$$

$$\dots$$

$$(n-1)n-1$$

Where m is impossible if we can write $m = qn + r$ for integers $q \geq 1$ and $q < r \leq n-1$.

Are impossible for a total of $(n-2) + (n-1) + \cdots + 1 = \frac{(n-1)(n-2)}{2}$ impossible.

We first prove that all numbers of the form $a(n+1) + b(n)$ work. We let an A-group of $n+1$ have colors $c_1, c_2, \dots, c_{n-1}, c_n, c_n$ where c_n is the only repeated. We let a B-group of n beads have colors $c_1, c_2, \dots, c_n$. One construction would be to have a A-groups back-to-back followed immediately by b B-groups back-to-back. (Again, wondering if I have to explain why this works?)

We prove no other m work. FTSOC let $m = qn + r$ for integers $q \geq 1$ and $q < r \leq n-1$ work. By Pigeonhole, there must be at least one color c_i that appears less than or equal to q times.

However, we may also write $m = q(n+1) + (r-q)$. Therefore, of the m ranges of $n+1$ colors (taking cyclically), there must be $r - q > 0$ without color c_i.

Problem 18 (13SLA4)

We prove that the sum of any optimized $\{a_1, \dots, a_n\}$ is the same as the sum n^2 of $\{n, n, \dots, n\}$ where optimized means the sequence a meets the requirements of the problem, but we cannot increase any term in a by one and still have it meet the requirements.

We prove this by construction. Let a_i be the biggest term of a less than or equal to n. In other words, i is the integer where $a_1, \dots, a_i \leq n; a_{i+1}, \dots, a_n > n$.

We let k be the greatest integer where $a_1 = a_2 = \cdots = a_k$.

Claim 1: $a_1 = a_2 = \cdots = a_k = i$

If $a_1 = \cdots = a_k = m$ for $m > i$, then we will need $a_{a_1} = a_m \leq n - 1 + 1$, but $a_m > n$.

If $a_1 = \cdots = a_k = m$ for $m < i$, then we show that $a_1 = \cdots = a_k = m + 1$ is also valid.

First, note that $a_{k+1} \neq m \Longrightarrow a_{k+1} > m \Longrightarrow a_{k+1} \geq m + 1$ so monotonic increasing is preserved. Since $m < i \Longrightarrow m + 1 \leq n$, the last condition is preserved as well.

Claim 2: Where $a_0 = 0$ for here, $a_{a_{j-1}+1} = \cdots = a_{a_j} = n + j - 1$ for all $1 \leq j \leq i + 1$

For $j \leq i$, this is obviously the upper-bound. If all of these were any lower, we use the bolded argument as before to see that it would not be optimal.

For $j = i + 1$, then this is the upper bound because $a_1, \dots, a_n \leq a_1 + n = i + 1 - 1$. If any lower, we use the bolded argument as before to see it would not be optimal.

We next check that $a_{a_j} \leq n + j - 1$ for all $1 \leq j \leq k$. We simply check the smallest bound,

$$a_{a_1} = a_{m+1} \leq a_i \leq n$$

Since $a_1 \dots a_k \leq i \leq n \leq n + j - 1$ for all $1 \leq j \leq n$, incrementing $a_1 = \cdots = a_j$ by one has will not upset the bound in any other way. (Does part make sense?)

Since we can increment $a_1 \dots a_k$ and still have a valid sequence, the initial sequence is not optimized so this case is impossible as well.

With the claim proven, we move to the construction. We fix i at the start of the construction as the integer where $a_1, \dots, a_i \leq n$ and $a_{i+1}, \dots, a_n > n$. We claim that, through a series of moves that increases one element of a by one and decreases another by one, we'll eventually get $a_1 = \cdots = a_i = n$.

The construction is as follows. We use pseudo-code to demonstrate this,

while $a_k \neq n$, {

Fix $m \coloneqq a_k$;

Increment a_k by one;

Decrease a_{a_k} by one (for the new a_k);

}

Since at every step of the construction, $m = i < a_m \leq n < a_{m+1}$, incrementing a_k by one and decreasing a_{m+1} by one will not break $a_1 \leq a_2 \leq \cdots \leq a_n \leq a_1 + n$.

Also, for the new a_k, a_{a_k} now must decrease by one as it was previously $n + k$ and now must be less than or equal to $n + k - 1$. All other terms may remain the same by the bolded argument.

Since the sum $a_1 + \cdots + a_k$ is monotonic increasing, after enough steps, $a_1 = \cdots = a_k = n$.

In a valid sequence where $a_1 = n$, then $a_2 = \cdots = a_n = n$ as since $a_{a_1} = a_n \leq n$. Therefore, we have proved that through a sequence of steps preserving the sum of the sequence, we have transformed our sequence into a sequence of all ns, therefore, the initial sum is less than or equal to n^2 and we are done.

<u>Feedback</u>

Brief feedback on the examples.

- For walkthrough of 1.6,
    - In part **a,** it's not too obvious to prove there are exactly seven 3's and 4's and each must be bitwise complement of another, so best to just say $n_i \in \{0, 3, 4, 7\}$
    - For the heart equation, maybe it's just me but coming from the previous problems I had a very hard time remembering that n_i is an integer. I think it's worth mentioning.
    - It suddenly jumps to saying vectors in part b
- Small typo in part 3 of solution 1.5. It should say "The **three** 'NE L-shape' partitions" and so on.

48 clubs; 19.5 hours